

PROJECT SCHEDULING – SHORT & LONG ANSWERS (≈75 WORDS EACH)

1. What is project scheduling?

Project scheduling is the process of planning project activities, determining their sequence, estimating durations, and assigning start and finish dates. It creates a structured timeline that guides execution, coordinates dependencies, and ensures tasks are completed logically. Effective scheduling helps manage time constraints, align resources, and achieve project objectives within planned deadlines.

2. Why is project scheduling important?

Project scheduling is important because it provides a clear roadmap for executing tasks on time. It helps avoid delays, improves coordination, enables progress tracking, and supports informed decision-making. A proper schedule ensures efficient use of resources and increases the likelihood of completing the project successfully within constraints.

3. Name two benefits of having a project schedule.

Two key benefits of a project schedule are better time management and improved task coordination. Scheduling helps teams understand deadlines and responsibilities while allowing managers to monitor progress, identify risks early, and take corrective actions to keep the project on track.

4. How does scheduling help in resource management?

Scheduling helps allocate resources at the right time and place. It prevents overuse or underuse of people, equipment, and materials. By aligning tasks with resource availability, scheduling improves efficiency, reduces conflicts, and ensures smooth project execution.

5. What challenges can arise if a project schedule is not well-defined?

A poorly defined schedule can cause confusion, missed deadlines, resource conflicts, and cost overruns. It makes progress tracking difficult and increases the risk of delays, rework, and stakeholder dissatisfaction.

6. What are project activities?

Project activities are individual tasks or work units required to complete project deliverables. Each activity consumes time and resources and contributes directly to achieving project objectives.

7. Role of WBS in defining activities?

WBS breaks the project into smaller deliverables, helping identify and define activities clearly. It improves planning accuracy and task management.

8. How do you define project activities?

Project activities are defined by decomposing deliverables into manageable tasks, clearly describing the work involved, and assigning responsibilities.

9. Relationship between project scope and activity definition?

Project scope defines what work is required, while activity definition specifies how that work will be completed through tasks.

10. Why clearly describe each task?

Clear task descriptions reduce confusion, improve accountability, and support accurate estimation of time and resources.

11. What does sequencing activities mean?

Sequencing activities means determining the logical order of tasks based on dependencies and

constraints.

12. Name four types of task dependencies.

Finish-to-Start, Start-to-Start, Finish-to-Finish, and Start-to-Finish dependencies.

13. What is Finish-to-Start dependency?

In Finish-to-Start dependency, one activity must finish before the next activity can start.

14. How does a network diagram help?

A network diagram visually shows activity relationships, helping identify dependencies and critical paths.

15. Difference between SS and FF dependencies?

SS allows activities to start together, while FF requires activities to finish together.

16. Three main types of resources?

Human resources, material resources, and equipment or financial resources.

17. Why is resource estimation important?

It ensures adequate resources are available, supports realistic scheduling, and prevents delays.

18. Factors influencing resource estimation?

Scope, complexity, availability, skills, technology, and budget constraints.

19. Effect of poor resource estimation?

It leads to delays, cost overruns, overworked teams, and reduced quality.

20. Two methods for estimating resources?

Analogous estimation and parametric estimation.

21. What is activity duration estimation?

It is the process of estimating the time required to complete each project activity.

22. Three techniques for estimating duration?

Analogous, parametric, and three-point estimation.

23. Purpose of three-point estimation?

It accounts for uncertainty by using optimistic, pessimistic, and most likely estimates.

24. Define optimistic, pessimistic, and most likely estimates.

Optimistic is best-case duration, pessimistic is worst-case, and most likely is realistic time.

25. Why is historical data useful?

It improves estimation accuracy using past project experience.

26. What is a Gantt Chart?

A Gantt Chart is a bar chart showing activities against time for tracking progress.

27. How does CPM help?

CPM identifies critical activities and minimum project duration.

28. Significance of critical path?

Delays on the critical path directly delay project completion.

29. What is Critical Chain Scheduling?

CCS focuses on resource constraints and uses buffers to protect schedules.

30. How does PERT help?

PERT manages uncertainty using probability-based duration estimates.

31. What is schedule control?

Schedule control involves monitoring progress and managing timeline changes.

32. Two tools for monitoring schedules?

Gantt charts and Earned Value Management.

33. What is schedule variance?

It is the difference between planned and actual schedule performance.

34. How can a manager handle delays?

By reallocating resources, revising schedules, and applying corrective actions.

35. Why track schedules regularly?

Regular tracking helps detect issues early and keep the project on track.

36. Two popular scheduling software?

Microsoft Project and Jira.

37. Advantage of scheduling software?

It automates planning, tracking, and reporting with better accuracy.

38. How software helps in resource allocation?

It visualizes resource usage and prevents over-allocation.

39. Limitations of scheduling software?

It requires accurate data, skilled users, and can be costly.

40. Why is training necessary?

Training ensures effective and accurate use of scheduling software.

LONG QUESTIONS

1. Importance of project scheduling and impact on success?

Project scheduling ensures timely completion, efficient resource use, and better coordination. It provides a clear roadmap, supports monitoring, reduces risks, and improves decision-making. A well-planned schedule increases the chances of meeting deadlines, controlling costs, and achieving project objectives successfully.

2. Benefits of a well-defined project schedule?

A well-defined schedule improves time control, resource planning, progress tracking, risk identification, and stakeholder communication, leading to successful project execution.

3. Scheduling in resource management and communication?

Scheduling aligns resources with timelines and provides clear information to stakeholders, improving coordination and transparency.

4. What is WBS and how it helps?

WBS breaks the project into smaller components, helping define activities clearly and manage work effectively.

5. Steps in defining activities?

Review scope, decompose deliverables, define tasks, document details, and assign responsibilities.

6. Contribution of activity list to success?

It improves clarity, accountability, estimation accuracy, and scheduling reliability.

7. Types of task dependencies with examples?

FS, SS, FF, and SF dependencies define how tasks relate to each other.

8. Role of network diagrams?

They visualize task sequences and help identify critical paths.

9. How sequencing avoids delays?

It ensures logical flow and prevents bottlenecks.

10. Types of project resources?

Human, material, equipment, and financial resources.

11. Factors influencing resource estimation?

Scope, complexity, skills, availability, and budget constraints.

12. Impact of improper estimation?

It causes delays, cost overruns, and quality issues.

13. Estimation techniques?

Analogous uses past data, parametric uses formulas, three-point uses ranges.

14. Role of historical data?

It improves accuracy by using past project experience.

15. Challenges in duration estimation?

Uncertainty, limited data, and changing requirements.

16. Explain CPM and significance?

CPM identifies critical activities affecting project duration and helps manage schedules.

17. Gantt Charts and example?

They show timelines visually, such as construction phase schedules.

18. CCS vs PERT?

CCS focuses on resource constraints, PERT on uncertainty.

19. Methods to control schedule?

Tracking, variance analysis, and corrective actions.

20. Handling unexpected delays?

By revising plans, reallocating resources, and applying risk responses.